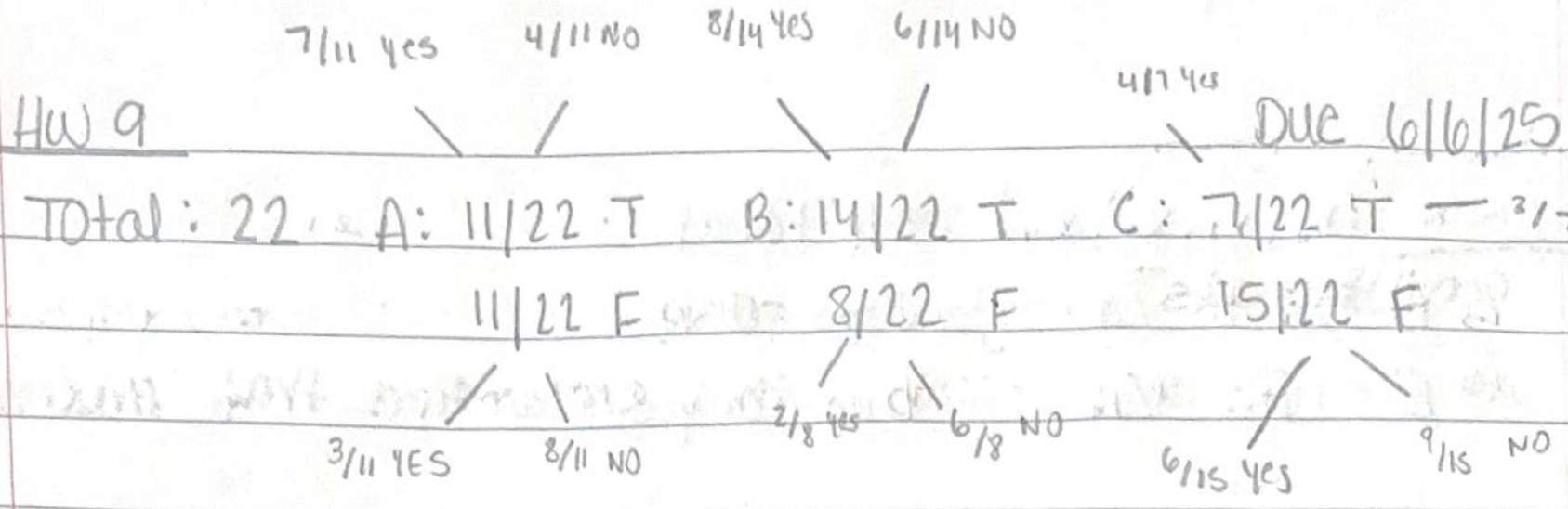
HW 9 — DUE 6/6/25



1) Total: 22 A: 11/22 T B: 14/22 T C: 7/22 T
11/22 F 8/22 F 15/22 F

A = T

$-\left(\frac{7}{11}\log_2 \frac{7}{11} + \frac{4}{11}\log_2 \frac{4}{11}\right) = 0.946$

A = F

$-\left(\frac{3}{11}\log_2 \frac{3}{11} + \frac{8}{11}\log_2 \frac{8}{11}\right) = 0.845$

A overall: $\frac{11}{22} \cdot 0.946 + \frac{11}{22} \cdot 0.845 = \boxed{0.8955}$

★ lowest cond entropy so split on A

B = T

$-\left(\frac{8}{14}\log_2 \frac{8}{14} + \frac{6}{14}\log_2 \frac{6}{14}\right) = 0.985$

B = F

$-\left(\frac{2}{8}\log_2 \frac{2}{8} + \frac{6}{8}\log_2 \frac{6}{8}\right) = 0.811$

B overall: $\frac{14}{22} \cdot 0.985 + \frac{8}{22} \cdot 0.811 = \boxed{0.922}$

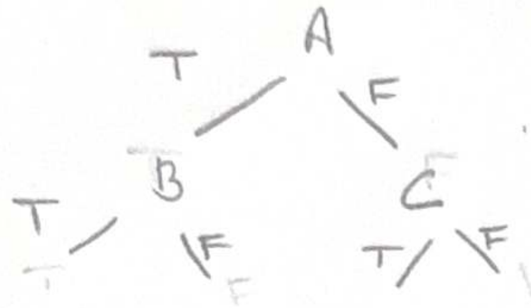
C = T

$-\left(\frac{4}{7}\log_2 \frac{4}{7} + \frac{3}{7}\log_2 \frac{3}{7}\right) = 0.985$

C = F

$-\left(\frac{6}{15}\log_2 \frac{6}{15} + \frac{9}{15}\log_2 \frac{9}{15}\right) = 0.971$

C overall: $\frac{7}{22} \cdot 0.985 + \frac{15}{22} \cdot 0.971 = \boxed{0.975}$



A=T $(x_1, x_2, x_3, x_4) \rightarrow$ 11 total

↳ B = T 7/11 7 yes 0 NO

↳ B = F 4/11 4 NO 0 yes

$H(B=t) = -(1 \cdot \log_2 1 + 0 \cdot \log_2 0) = 0$

$H(B=f) = -(0 \cdot \log_2 0 + 1 \cdot \log_2 1) = 0$

$H(D|B, A=t) = \frac{7}{11} \cdot 0 + \frac{4}{11} \cdot 0 = 0$

↳ C = T 4/11 1 yes 3 NO

↳ C = F 7/11 6 yes 1 NO

$H(C=t) = -\left(\frac{1}{4} \log_2 \frac{1}{4} + \frac{3}{4} \log_2 \frac{3}{4}\right) = 0.811$

$H(C=f) = -\left(\frac{6}{7} \log_2 \frac{6}{7} + \frac{1}{7} \log_2 \frac{1}{7}\right) = 0.592$

$H(D|C, A=t) = \frac{4}{11}(0.811) + \frac{7}{11}(0.592) = 0.672$

A = true

0 < 0.672 so 2nd branch on B, lower entropy

A=F $(x_5, x_6, x_7, x_8) \rightarrow$ 11 total

↳ B = T 7/11 1 yes 6 NO

↳ B = F 4/11 2 yes 2 NO

$H(B=t) = -\left(\frac{1}{7} \log_2 \frac{1}{7} + \frac{6}{7} \log_2 \frac{6}{7}\right) = 0.592$

$H(B=f) = -\left(\frac{2}{4} \log_2 \frac{2}{4} + \frac{2}{4} \log_2 \frac{2}{4}\right) = 1$

$H(D|B, A=f) = \frac{7}{11} \cdot 0.592 + \frac{4}{11} \cdot 1 = 0.740$

↳ C = t 3/11 3 yes · 0 NO

↳ C = f 8/11 8 NO · 0 yes

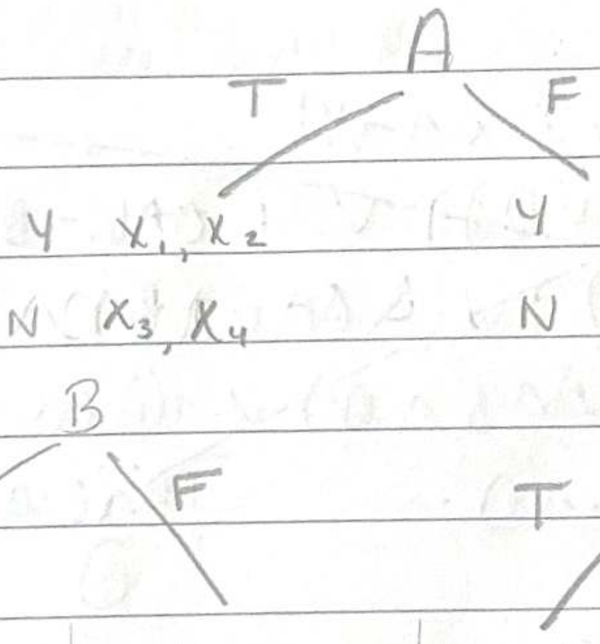
H(C=t) = 0

H(C=f) = 0

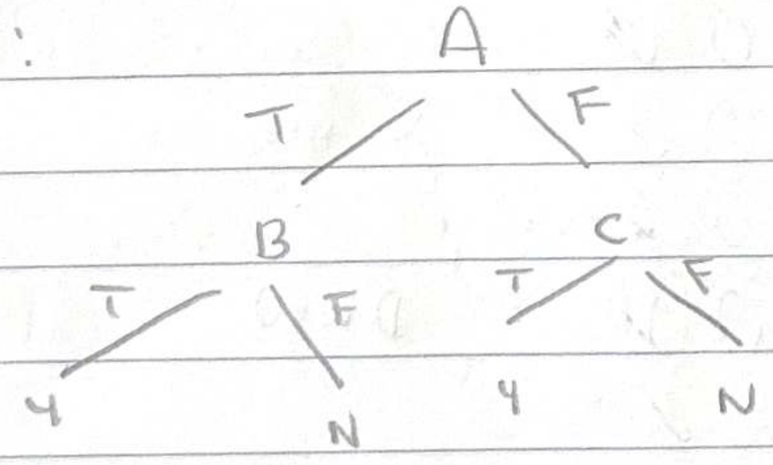
H(D | C, A=F) = 3/11 · 0 + 8/11 · 0 = (0)

0 < 0.740 so cond on C in F branch. b/c lower entropy

A =



Final:



$\oplus$

2) $(A \vee \neg B)$ XOR $(\neg C \vee D)$ $\quad X \oplus Y = (X \vee Y) \wedge (\neg X \vee \neg Y)$

- only 1 can be True or False $\quad = (X \wedge \neg Y) \vee (\neg X \wedge Y)$

* chose and option

$= [(A \vee \neg B) \wedge \neg(\neg C \vee D)] \vee [\neg(A \vee \neg B) \wedge (\neg C \vee D)]$

$= [(A \vee \neg B) \wedge (C \wedge \neg D)] \vee [(\neg A \wedge B) \wedge (\neg C \vee D)]$

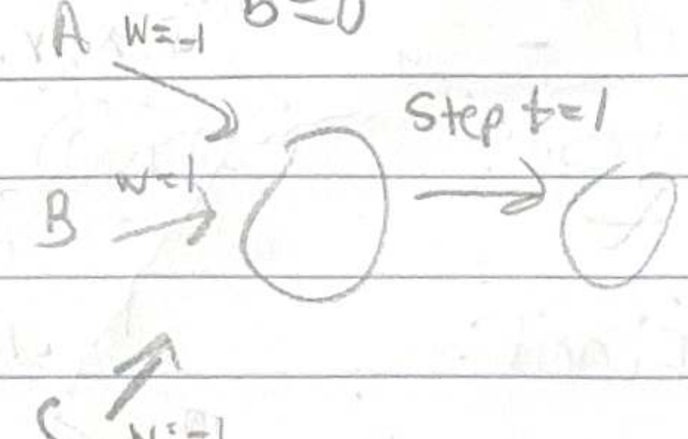
$[(A \vee \neg B) \wedge C \wedge \neg D] \vee [\neg A \wedge B \wedge (\neg C \vee D)]$

$(A \wedge C \wedge \neg D) \vee (\neg B \wedge C \wedge \neg D) \vee (\neg A \wedge B \wedge \neg C) \vee (\neg A \wedge B \wedge D)$

(3) (4) (1) (2)

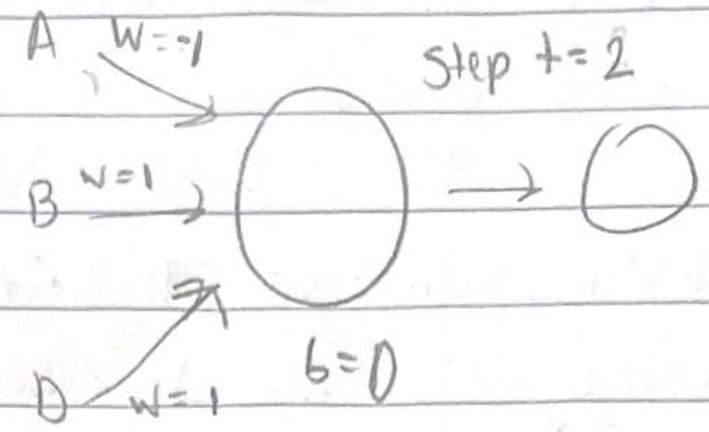
(1)

A	B	C	¬A	¬C	¬A ∧ B ∧ ¬C
0	0	1	1	0	0 ✓
0	1	0	1	1	1 ✓
0	1	1	1	0	0 ✓
0	0	0	1	1	0 ✓
1	1	1	0	0	0 ✓
1	0	0	0	1	0 ✓
1	1	0	0	1	0 ✓
1	0	1	0	0	0 ✓



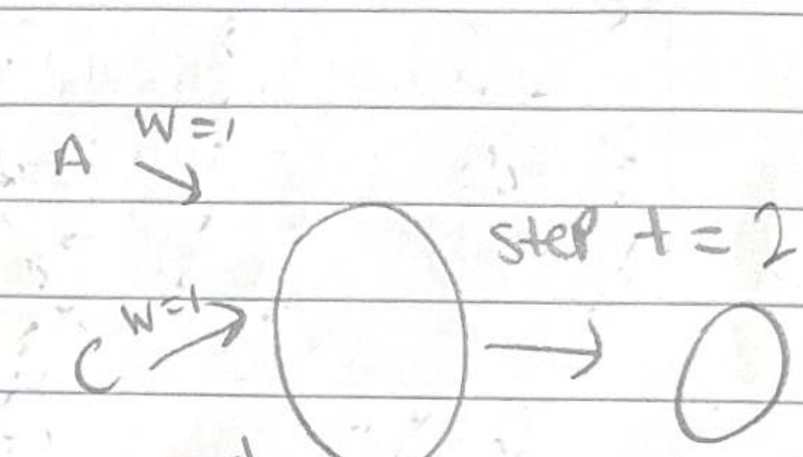
(2) $\neg A \wedge B \wedge D$

A	¬A	B	D	¬A∧B∧D
0	1	0	0	0 ✓
0	1	1	0	0 ✓
0	1	0	1	0 ✓
0	1	1	1	1 ✓
1	0	1	1	0 ✓
1	0	0	0	0 ✓
1	0	1	0	0 ✓
1	0	0	1	0 ✓



(3) $A \wedge C \wedge \neg D$

A	C	D	¬D	A∧C∧¬D
0	0	0	1	0 ✓
0	1	0	1	0 ✓
0	0	1	0	0 ✓
0	1	1	0	0 ✓
1	1	1	0	0 ✓
1	0	0	1	0 ✓
1	0	1	0	0 ✓
1	1	0	1	1 ✓

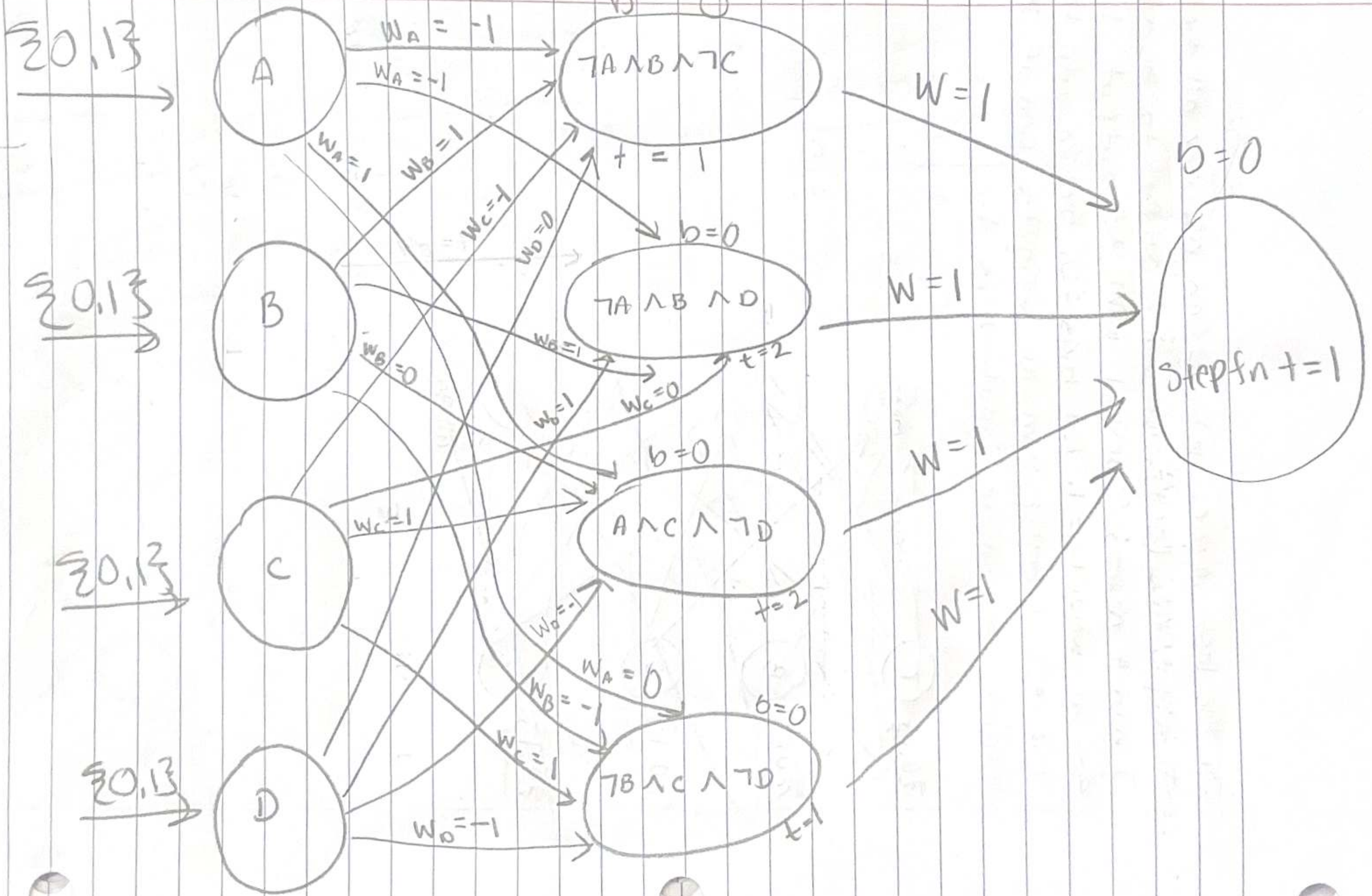


(4) $\neg B \wedge C \wedge \neg D$

B	¬B	C	D	¬D	¬B∧C∧¬D
0	1	0	0	1	0 ✓
0	1	1	0	1	1 ✓
0	1	0	1	0	0 ✓
0	1	1	1	0	0 ✓
1	0	0	0	1	0 ✓
1	0	1	1	0	0 ✓
1	0	1	0	1	0 ✓
1	0	0	1	0	0 ✓

For the last layer, we have an OR over all the last nodes.
If any is true (or 1), we want output to be yes.
I had # nodes; if any = 1 want to output 1.
Set all weights = 1. Let bias = 0. Step fn with t=1.
We essentially sum all weights and if ≥1, that's means ≥1 node inputted is 1.

all step fn
w/ diff ts



3) a) True. Each layer uses a linear function, so the overall result is a combination of linear results. Non-linearity is never introduced.

b) False. Gradient descent is not guaranteed to converge when the function is not convex.
to a global minimum

c) False. Neural networks can do both supervised and unsupervised learning. In unsupervised learning, labeled data is not needed, so they can be trained on unlabeled data.